

SECTION 2 - DOMESTIC MAGNET DELIVERY

2-1 IN-TRANSIT HANDLING AND SERVICE

TABLE 2-1
LIFTING / HANDLING INFORMATION

MAGNET	MAXIMUM WEIGHT ^a	MAXIMUM TILT ^b	FORKLIFT CAPABILITY
LCC 1.5T	12,500 pounds (5,670 kg)	30 degrees	Yes
LCC 1.0T			
^a Includes 3000 pounds (1360 kg) weight of the gradient coil and 126 pounds (58 kg) for associated coil hardware, including cryogens. ^b Maximum tilt allowed only when supported by feet for forklift / rigging operations or moving on inclines (base lift).			



The magnet CANNOT be shipped by train as serious damage to the magnet's internal suspension system may occur due to the vibration loads introduced by rail systems.

TABLE 2-2
SHIPPING INFORMATION

MAGNET	ALLOWABLE SHIPPING MODES	SHIPPING CAPABILITY	MAXIMUM TRANSIT TIME	COMMENTS
LCC 1.5T	Airplane, air ride only trailer, boat or ocean-going ship	Warm or Cold	Cryogen refill for cold shipment = 15 days	Only tie downs to magnet feet and lifting rails are permitted.
LCC 1.0T				

TABLE 2-3
MAXIMUM LOAD INFORMATION

MAGNET	MAXIMUM LONGITUDINAL LOAD	MAXIMUM LATERAL LOAD	MAXIMUM VERTICAL LOAD	MAXIMUM SHOCK LOADS
LCC 1.5T	1.2 G	1.5 G	1.5 G	None Allowed
LCC 1.0T				

Note

Only tie downs to magnet feet and lifting rails are permitted.

2-1-1 In-Transit Helium Fill

WARNING!

THIS PROCEDURE MUST BE PERFORMED BY QUALIFIED PERSONNEL ONLY. MAKE SURE THE SAFETY PRECAUTIONS LISTED BELOW ARE COMPLETELY FOLLOWED WHEN PERFORMING ANY SERVICE THAT WILL RESULT IN HELIUM EXHAUST FROM THE MAGNET.

LARGE QUANTITIES OF HELIUM GAS WILL DISPLACE OXYGEN IN A CLOSED ROOM. FOR YOUR PERSONAL SAFETY, MAKE SURE THE MAGNET STORAGE AREA IS ADEQUATELY VENTILATED BY OPENING DOORS AND WINDOWS. THE ADDITION OF PORTABLE FANS MAY BE NECESSARY IN CASES OF SMALL ROOM SIZE WITH POOR VENTILATION. THIS IS REQUIRED.

SKIN CONTACT WITH LIQUID CRYOGENS WILL CAUSE BURNS. WEAR PROTECTIVE CLOTHING, GLOVES (NONABSORBENT MATERIAL) AND GOGGLES OR FACE SHIELD WHEN TRANSFERRING CRYOGENS. MAKE SURE A SECOND PERSON IS PRESENT WHILE SERVICING THE MAGNET IN CASE OF ANY EMERGENCY.

SMOKING IS PROHIBITED IN THE MAGNET AND CRYOGEN STORAGE ROOMS. LIQUID CRYOGENS CAN LIQUIFY ATMOSPHERIC OXYGEN THUS PRODUCING A HIGHLY ENRICHED OXYGEN LIQUID.

2-2 EQUIPMENT / TOOLS FOR MAGNET INSTALLATION

2-2-1 Forklift

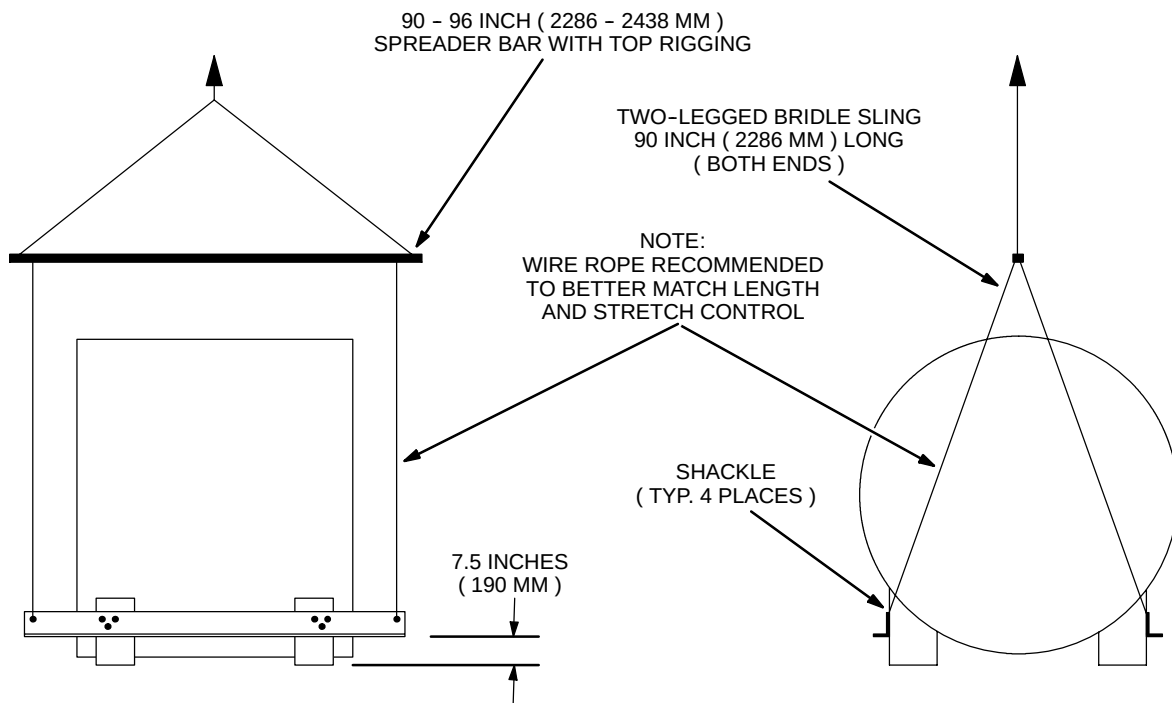
TABLE 2-4
FORKLIFT REQUIREMENTS

ITEM	QTY.	SPECIFICATIONS		FURNISHED BY:	INTENDED USE:
Forklift	1	Forklift Capacity:	12,500 pounds (5,670 kg) at 50 inches (1,270 mm) to load center	Rigger	Unloading or moving magnet (without pallet)
		Minimum Fork Length:	90 inches (2,286 mm)		
		Minimum Distance Between Forks:	80 inches (2,032 mm)		

2-2-2 Crane Lift

TABLE 2-5
CRANE REQUIREMENTS

ITEM	QTY.	SPECIFICATIONS		FURNISHED BY	INTENDED USE
Crane	1	Total Load:	12,500 pounds (5,670 kg)	Rigger	Unloading or moving magnet (without pallet)
Spreader Beam	1	Total Load:	12,500 pounds (5,670 kg)	Rigger	Unloading or moving magnet (without pallet)
		Distance Between Lift Points:	90 - 96 inches (1,219 - 1,270 mm)		
Two Legged Cable Bridle	2	Total Load:	12,500 pounds (5,670 kg)	Rigger	Unloading or moving magnet (without pallet)
		Leg Length:	90 inches (2286 mm) min. (all legs same length)		
Shackle	4	Pin Diameter:	1 inch (25 mm)	Rigger	Unloading or moving magnet (without pallet)
		Minimum Work Load Limit:	6,500 pounds / shackle		

SPREADER BEAM AND BRIDLE SLING
ILLUSTRATION 2-1

2-2-3 Miscellaneous Equipment / Tools

TABLE 2-6
MISCELLANEOUS EQUIPMENT / TOOLS

ITEM	QTY.	EQUIPMENT / TOOL REQUIREMENT	RESPONSIBLE	FUNCTION
Hydraulic or Toe Jack	4	Must support one end of magnet on two jacks or both ends of magnet on 4 jacks Total weight: 12,300 pounds (5,580 kg)	Rigger	Raise magnet for roller dollies or leveling plates
Roller Dollies	4	Total weight for all dollies combined: 4,100 pounds (1,860 kg)	Rigger	Moving magnet to magnet room
Level	1	24 - 36 inch (610 - 915 mm) length	Rigger	Level the magnet
Magnet Leveling Kit	1	N / A	GE / SYS	46-260888G4 Level the magnet
Magnet Bolt-down / Seismic Kit	1	Apply sufficient torque to provide 2,500 pounds (1,134 Kg) clamping force per anchor. Torque requirements to be supplied by screen room vendor.	Screen Room Vendor	Bolt down magnet
Torque Wrench and Socket	1	Torque range: 58 to 258 ft-lbs. Deep socket, size dependant on anchor selection	Rigger	Bolt down magnet
Socket Wrench and Sockets	1	3/4 inch socket (add sockets for crate removal)	Rigger	Remove 1/2 inch crate lag screws
Magnet Interface Drawing	1	N / A	GE / ISS	22272411DW Identify magnet dimensions & features
Pre-Delivery Information Package	1	N / A	GE	Aid in magnet installation
Portable Temperature Meter	1	N / A	GE / FE	2125073 / Check internal temperatures
Digital Volt / Ohm Meter	1	Fluke model or equivalent	GE / FE	Perform magnet electrical checks
Magnet System Components Checks: Tables 3-1 to 3-5	1	N / A	GE / SYS	Confirm electrical check values
Wood Timber	2	4 inch x 6 inch x 8 foot	Rigger	Moving / positioning magnet to magnet room
Steel or Aluminum Plate	6	18 inch x 6 foot, .25 inch thick	Rigger	Moving magnet to magnet room
Pry-bar	2	Length as required to move magnet	Rigger	Moving / positioning magnet in magnet room
RuO Temperature Meter	1	N / A	GE	Check second stage & recondenser temperatures
Adapter Cable for Diode Temperature Meter	1	N / A	GE	Check internal temperatures
Circular or Chain Saw	AR	N / A	Rigger	For cutting wood timbers
Heavy Duty Drill & Drill Bits	AR	N / A	Rigger	To add holes to wood timbers
Min. .50 inch dia. Mounting Hardware	AR	N / A	Rigger	For attaching wood timbers to rails

2-3 CRATE AND PROTECTIVE PACKAGING REMOVAL



Care must be taken not to scrape or hit the sides of the magnet.

The magnet is shipped with bubble-wrap plastic wrap and a separate outer plastic bag. The bubble-wrap should be left intact until the magnet is anchored in the screen room. Care must be taken in removing the plastic bag to avoid damage to various magnet components.

1. The crate can be removed with crane either by lifting straight up or by removing the bolted boards on one end and pulling the crate toward that end.
2. The outer plastic bag should then be removed.



MAGNET AS SHIPPED (PLASTIC BAG AND BUBBLE-WRAP)
ILLUSTRATION 2-2

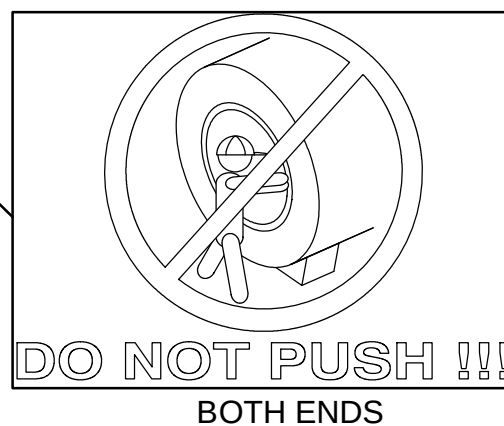
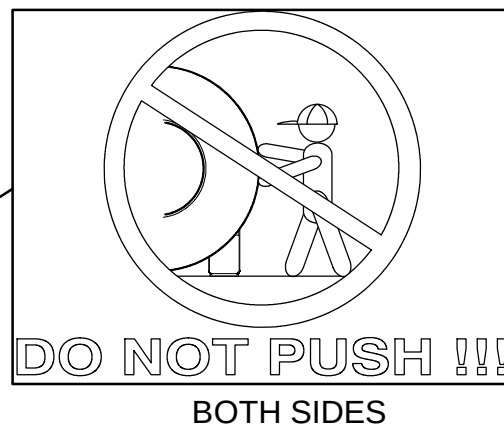
2-4 UNLOADING MAGNET

WARNING!

DO NOT APPLY ANY FORCE TO THE FIBERGLASS COVERS ON THE MAGNET'S TOP, SIDES OR ENDS (SEE ILLUSTRATION 2-3).

USE LIFTING CABLES AS SHOWN TO LIFT MAGNET.

BEFORE MOVING THE MAGNET TO THE MR SUITE, REFER TO SECTION 5.



MAGNET NO-PUSH SIGN LOCATIONS
ILLUSTRATION 2-3

2-4-1 Forklift Method



Extreme care must be exercised during forklift operations:

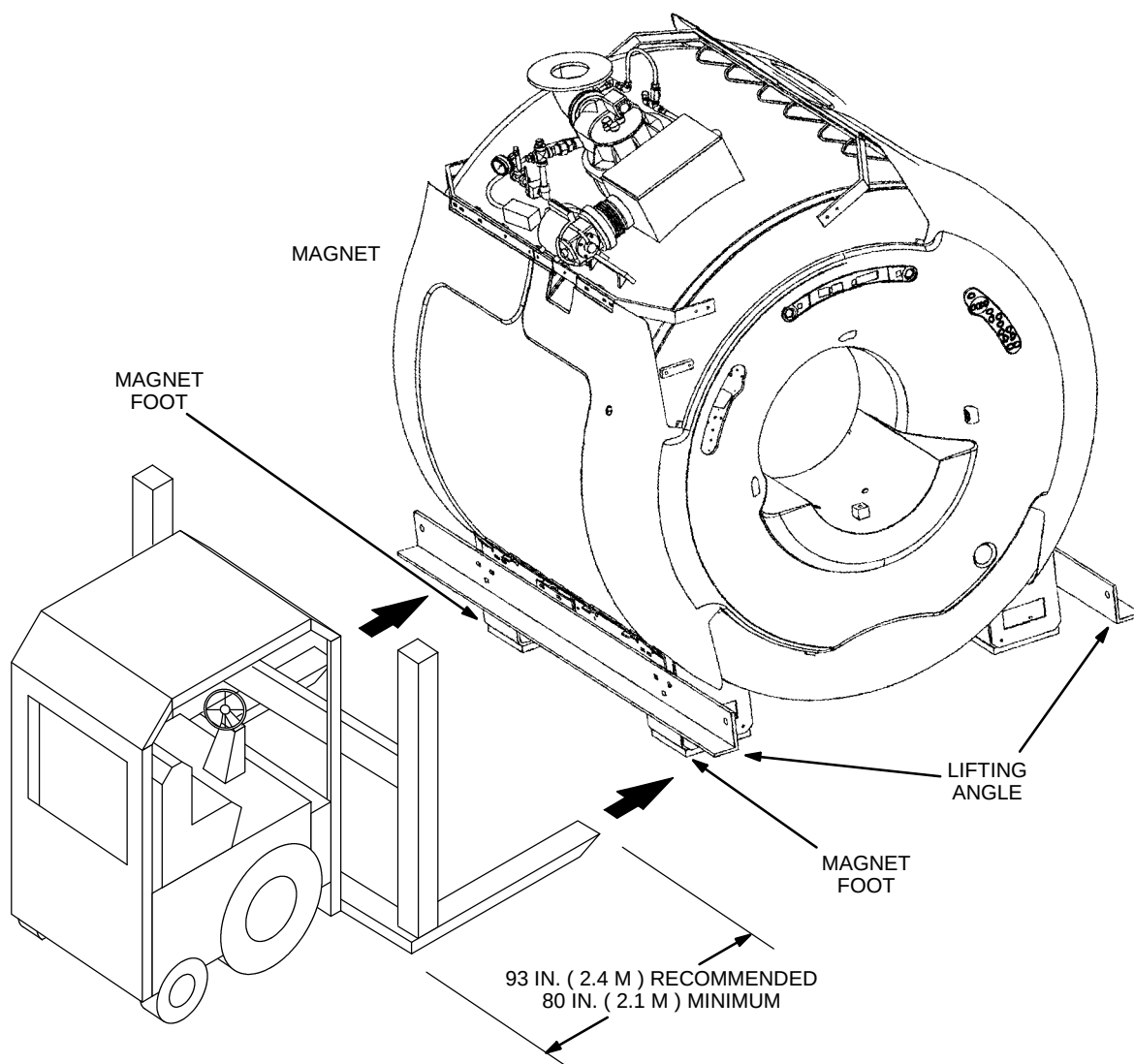
- S** It is important to use the proper size forklift. Make sure the forklift capacity is rated for the weight of the crate and the contents. See Illustration 2–4 for required fork dimensions. The minimum fork length is 2.1 Meters (90 inches).
- S** The magnet **MUST** be picked up from the sides only. The forks must be placed under areas indicated on lifting angles.
- S** The magnet **MUST** be lifted smoothly to avoid impact or jolts to the system which may cause damage to the magnet.

1. Position the forklift at the side of the magnet and facing the magnet. Locate the forks under the lifting angles in the areas indicated. See Illustrations 2-4.
2. Carefully drive the forklift until forks are completely under both lifting angles in the areas indicated.
3. Lift magnet with the forklift and move smoothly to the desired location.

Alternate Magnet Moving Methods

Refer to Appendix A.

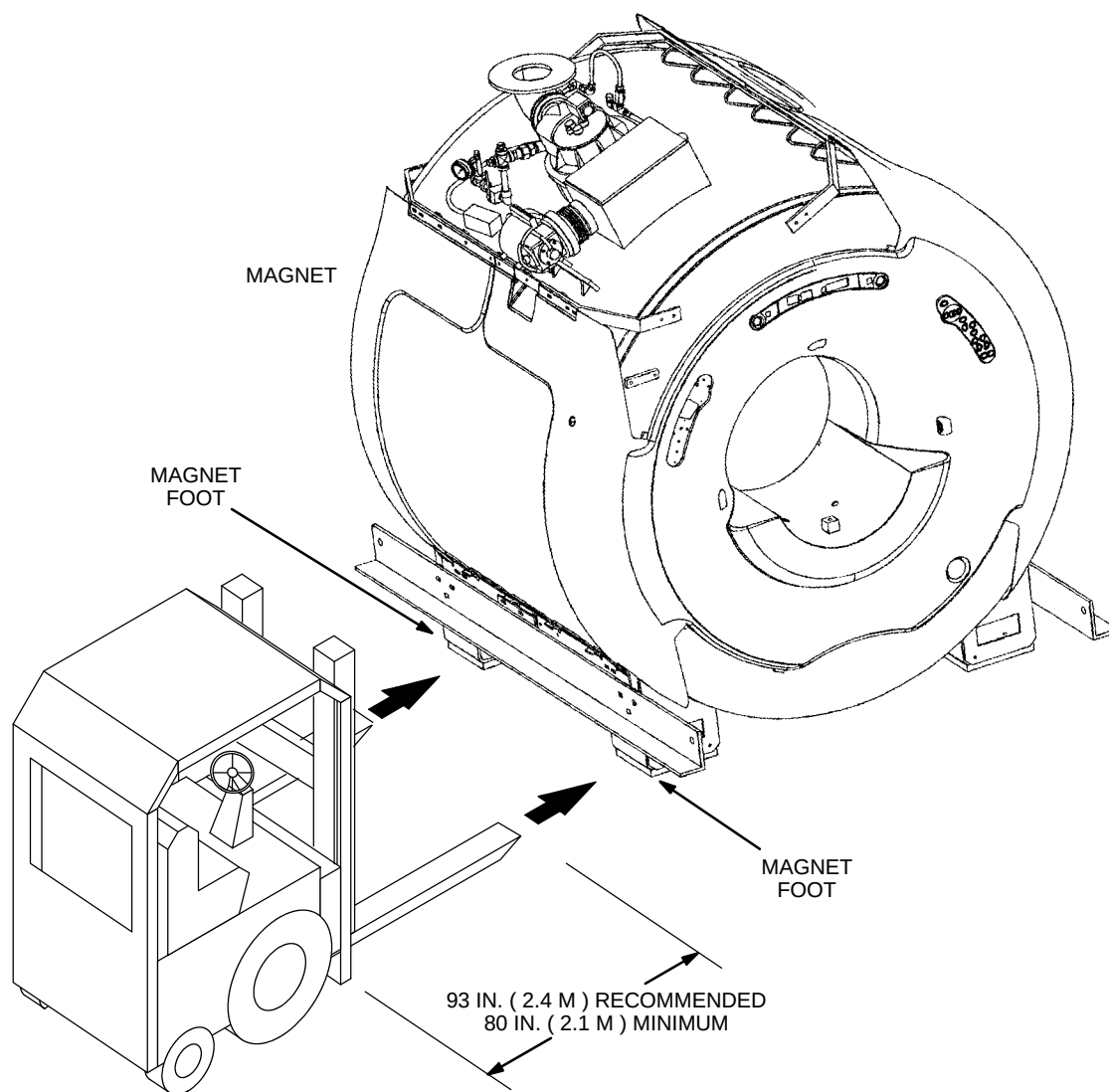
2-4-1 Forklift Method (continued)



NOTE:
FORKS MUST BE POSITIONED UNDER BOTH ENDS OF
LIFTING RAILS JUST OUTSIDE OF MAGNET FEET.

FORKLIFTING UNDER LIFTING RAILS
ILLUSTRATION 2-4

2-4-1 Forklift Method (continued)



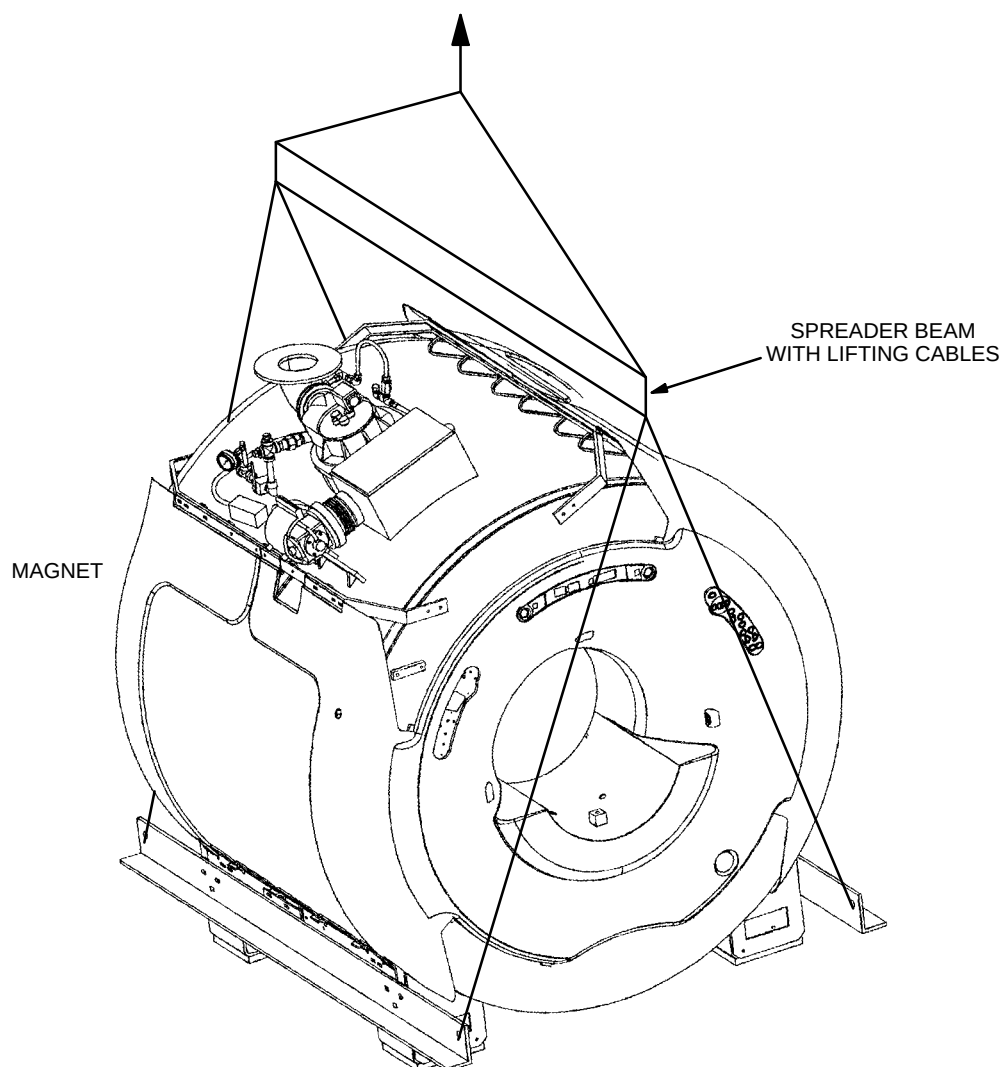
NOTE:
FORKS MUST BE POSITIONED UNDER BOTH MAGNET FEET.

FORKLIFTING UNDER MAGNET FEET
ILLUSTRATION 2-5

2-4-2 Crane Method



The lifting hooks of the spreader beam MUST be located as shown in Illustration 2-7. The magnet can be lifted by a crane which meets the specifications listed in Table 2-5. Failure to follow these requirements can result in serious damage to the magnet.



ALSO SEE PHOTO IN ILLUSTRATION 2-7.

LIFTING CONFIGURATIONS FOR CRANE
ILLUSTRATION 2-6

2-4-2 Crane Method (continued)



It is important to lift the magnet smoothly to avoid impact or jolts to the system. If severe jolting is encountered, internal damage may result.

When lowering the magnet, be careful not to shock the magnet, which could result in over stressing the internal magnet support members.

1. Position the crane hook centrally over the magnet to ensure a vertical lifting force on the lifting cables. See Illustration 2-7.
2. Attach the rigging to the lifting cables at each end of the magnet. See Illustration 2-7.
3. Lift the magnet with the crane.
4. Carefully lower the magnet onto a flat surface.



MAGNET LIFTING STRAP LOCATIONS
ILLUSTRATION 2-7

